

FIRM ENTRY AND EXIT DURING COVID-19: ADMINISTRATIVE EVIDENCE FROM GEORGIA

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Abstract

We use administrative business-register data from Georgia to examine how Small and Medium-Sized Enterprises (SMEs) adjusted during COVID-19. Covering 366,979 enterprises, the dataset allows us to construct monthly regional measures of firm openings, firm closings, net firm creation, and business entry rates. We combine these data with regional COVID-19 mortality and satellite-based proxies for local economic conditions over March 2020 to November 2022. SME activity fell sharply during the pandemic, especially during the first lockdown. The clearest disruption appears on the entry margin: higher mortality is associated with lower SME opening rates once regional sector composition is taken into account, and this relationship is concentrated in small rather than medium firms. The findings suggest that crisis policy should pay particular attention to preserving entry among small firms, where pandemic disruption appears strongest and where support may help protect more fragile entrepreneurial activity.

JEL codes: D22, L25, L26, H12, I18

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1 Introduction

The COVID-19 pandemic and the restrictions adopted to contain it disrupted business activity around the world. Small and Medium-Sized Enterprises (SMEs) were especially exposed, both because they account for a large share of firms and employment and because they often operate with limited liquidity, thinner buffers, and less scope to absorb sudden shocks (Thorgren and Williams, 2020). In that environment, understanding how crises affect firm entry, firm exit, and the net creation of firms is important not only for measuring short-run business disruption, but also for understanding how entrepreneurial activity adjusts when uncertainty rises and operating conditions deteriorate.

Much of the early evidence on COVID-19 and SMEs came from rapid-response surveys documenting cash-flow stress, payment deferrals, temporary closure, and sharp reductions in activity (Apedo-Amah et al., 2020; Bartik et al., 2020; Cowling et al., 2020; Cirera et al., 2021; Fairlie et al., 2023). That literature has been highly informative about short-run business stress, but it is less suited to tracing realized firm demography over time because it typically relies on self-reported outcomes and selected samples. We still know much less about how the pandemic affected realized firm entry, firm exit, and net firm creation in administrative data, especially at monthly frequency and below the national level.

We study these questions using administrative business-register data from Georgia. The Georgian Business Register gives near-universe coverage of registered SMEs and allows us to construct monthly regional measures of firm openings, firm closings, net firm creation, and entry rates from observed rather than self-reported outcomes. We combine those data with regional COVID-19 mortality and satellite-based nightlights, which provide a monthly proxy for local economic conditions. Georgia is a useful setting for this exercise because it combines a strict first-wave lockdown, later reopening, and substantial regional heterogeneity in pandemic intensity and economic conditions. Empirically, we exploit within-region variation in pandemic intensity in a region-month panel with region fixed effects and month-by-date fixed effects.

The main findings reveal a marked asymmetry between firm entry and exit. Descriptively, national openings and entry rates fell sharply during the first lockdown, with the contraction concentrated among small firms and in service-oriented sectors. In the panel regressions, higher COVID-19 mortality is associated with lower opening rates, especially once sector composition is taken into account, and the size-split results show that this relationship is concentrated among small rather than medium firms. By contrast, we find no corresponding increase in formal closures: the estimated closure-rate responses remain close to zero and statistically imprecise. The estimates for net firm creation are also weaker under small-cluster inference. Robustness exercises preserve the negative sign of the small-firm entry result, while placebo exercises do not indicate spurious timing. Taken together, the results suggest that greater pandemic intensity was more immediately reflected in missing entrants than in the formal exit of incumbent firms.

This weak closure response is consistent with recent evidence that the COVID-19 recession did not trigger

a large increase in firm exit. Kozeniauskas et al. (2022) find no rise in exit among lower-productivity firms in Portugal, while Banerjee and Ćirjaković (2026) show that firm exit did not increase in Slovenia during the pandemic. Relatedly, Gourinchas et al. (2025) estimate that SME failure rates in Europe would have risen substantially in the absence of government support, suggesting that policy interventions muted extensive-margin adjustment. Evidence from developing countries also points to resilience among incumbent formal firms: Bachas et al. (2025) find that realized 2020 outcomes were considerably stronger than predicted, partly because firms adjusted costs more flexibly than expected and large firms proved especially resilient.

Our findings also complement those of Bahaj et al. (2024), who document an increase in firm creation during COVID-19 in the UK. In contrast, we find that greater pandemic intensity reduced firm entry in Georgia, particularly among small firms, highlighting that the entrepreneurial response to COVID-19 differed substantially across contexts.

Several mechanisms can rationalize why the clearest response appears in openings rather than closures. Starting a firm requires making upfront commitments before revenues are realized and before future demand is fully known, arranging finance, searching for inputs or workers, and completing registration. When local health risk rises, these fixed and preparatory costs become more salient, so prospective entrepreneurs can respond by waiting. Incumbents face a different adjustment problem: they may reduce hours, scale down activity, draw on buffers, renegotiate obligations, or remain legally registered even when operations are temporarily depressed. The register therefore captures a shock that is more immediately visible in missing entrants than in formal exits.

To formalize one dimension of this asymmetry, we develop a simple theoretical framework with heterogeneous firms, endogenous scale, and selection into entry and continued operation. Potential entrants can avoid paying a sunk entry cost by waiting, whereas incumbents have already incurred that cost and operate with installed capital. A temporary decline in aggregate demand raises the productivity cutoff for entry, disproportionately deterring firms that would have operated with fewer workers. Fixed entry and operating costs also select more productive firms into incumbency, allowing a subsequent intensification of the pandemic to reduce entry without necessarily generating additional exit. The model additionally assumes that incumbents choose capital in advance. This planning lag distinguishes the unexpected arrival of COVID-19 from its subsequent intensification and allows some firms to exit initially even though surviving incumbents remain active when the pandemic becomes more severe.

The framework is related to models emphasizing financial constraints, short-run adjustment, and selection during recessions. Ayres and Raveendranathan (2023) show that tighter collateral constraints discourage entry and increase exit by restricting firms' ability to finance capital, while Gourinchas et al. (2025) model COVID-19 as a liquidity shock that may induce financially constrained firms to mothball production or fail. Kozeniauskas et al. (2022) emphasize how government support can weaken the usual selection of less productive firms through exit, and Bachas et al. (2025) document substantial flexibility in incumbent firms' adjustment of labor and

other costs. Our model combines related insights to explain a distinct asymmetry: after the initial selection and capital adjustment of incumbents, worsening pandemic conditions can continue to deter entry, particularly among potential small firms, without producing a corresponding increase in formal closures.

The policy relevance of this paper follows from this distinction between stabilizing incumbents and sustaining the pipeline of new firms. Liquidity support and job-retention schemes can prevent avoidable closures, but they need not relax the financing, planning, and search constraints faced by potential entrants (Eichhorst et al., 2022). Evidence from Portugal and Estonia suggests that COVID-era support policies muted productivity-enhancing reallocation and weakened adjustment on the exit or employment-reallocation margins (Kozeniauskas et al., 2022; Meriküll and Paulus, 2024). In Georgia, the concentration of the response among small-firm openings suggests that this distinction is especially important for low-buffer ventures, which face tighter financing constraints and greater fragility under low-resource conditions (Braggion et al., 2018; Morris et al., 2023). The Georgian evidence therefore shifts the discussion from reported distress to realized business formation and suggests that crisis policy should consider not only how to preserve existing firms, but also how to keep entry feasible when uncertainty and operating frictions are high.

The paper contributes in three ways. First, it provides population-level administrative evidence on SME dynamics during COVID-19 using near-universe administrative data. Second, it shows that the pandemic primarily disrupted firm entry rather than firm exit, with effects concentrated among small firms, highlighting an important asymmetry in how entrepreneurial activity adjusted during the crisis. Third, it provides evidence from a developing economy, complementing the emerging evidence from advanced economies and showing that the entrepreneurial response to COVID-19 was far from uniform.

The remainder of the paper is organized as follows. Section 2 outlines Georgia’s COVID-19 policies. Section 3 describes the data. Section 4 presents the empirical methodology. Section 5 reports results on SME openings, closures, and net firm creation. Section 6 presents a simple model that rationalizes the mechanism behind the asymmetric responses of firm entry and exit. Section 7 concludes with policy implications.

2 Background

The COVID-19 pandemic in Georgia unfolded across three distinct phases during 2020, each characterized by different policy approaches and economic impacts. The initial outbreak phase began on February 26, 2020, with Georgia’s first confirmed case. The government responded quickly with preventive measures including advancing school holidays and restricting public gatherings. To support businesses through the early disruption, it introduced tax benefits, notably four-month deferrals for the tourism sector, while commercial banks offered loan repayment postponements.

The second phase involved a tightening of restrictions as confirmed cases increased to 38 by March 18. A state of emergency was declared on March 21, accompanied by a nationwide curfew (9 PM to 6 AM) and

the closure of non-essential businesses. Only grocery stores, pharmacies, banks, gas stations, and post offices remained operational. International travel was suspended, and returning citizens were required to undergo a two-week quarantine. Economic support expanded with a comprehensive aid package: individuals who lost employment received 200 Georgian lari (GEL) per month for six months (roughly USD 65 at the 2020 average exchange rate), while self-employed persons received a one-time payment of 300 GEL (about USD 95). By April 14, with cases reaching 300, authorities further tightened measures by restricting entry and exit in major cities including Tbilisi, Kutaisi, Batumi, and Rustavi.

The final phase marked Georgia’s gradual reopening from late April onward, balancing economic recovery with ongoing public health concerns. Vehicle movement resumed nationwide, and agricultural markets reopened. In May, the government implemented a multi-stage economic revival plan, starting with construction and automotive services. By June, most economic activities had resumed, including domestic tourism (June 15) and railway transportation. Georgia’s effective management of the first wave resulted in relatively low infection rates – only 17 deaths by August 10, 2020 – and, according to contemporaneous assessments by the Deep Knowledge Group, ranked 39th among 200 nations in COVID-19 preparedness and response.

A comprehensive chronological timeline of policy measures and major COVID-19 developments in Georgia over our study period is provided in Table 1.

3 Data

3.1 Data Sources

We draw on a unique dataset assembled through special access to the Business Register of the National Statistics Office of Georgia to investigate SME dynamics during and after the COVID-19 pandemic. This access provides comprehensive firm-level information that is not publicly available and offers unusually broad coverage of the Georgian business landscape. Our analysis spans the years 2018 to 2023, allowing us to contrast the pre-pandemic period (2018–2019) with the pandemic and early recovery years (2020–2023).

The Business Register captures the full population of enterprise groups, enterprises, and local units active in the Georgian economy. From that source, we construct the SME dataset used in this paper. The database includes 366,991 SMEs with detailed identification, demographic, and economic characteristics. After excluding 12 firms registered in occupied regions where reliable data are unavailable, the final dataset comprises 366,979 enterprises across 11 regions, providing near-universe coverage of registered SMEs in Georgia.

In addition to the Business Register, we use two complementary data sources. First, we use monthly regional deaths by cause for 2020–2022, provided by the National Statistics Office of Georgia (NatStat Georgia), to isolate COVID-19 fatalities and measure the direct health burden of the pandemic across regions. Second, we incorporate satellite-based nightlight emissions from the dataset developed by Hu and Yao (2022), covering

2013–2023. These data provide a widely used proxy for regional economic activity and help capture local fluctuations that may not be fully reflected in official statistics.

We merge these three sources into a unified enterprise–region–month panel dataset. From the Business Register, we extract variables on firm dynamics (openings, closures, operational status, age, size, sector, and location). These are complemented by regional health and demographic indicators (COVID-19 deaths, [overall](#) mortality rates, and population counts), which capture the direct health burden of the pandemic. Finally, satellite-based nightlight intensity serves as a monthly proxy for regional conditions. Following Hu and Yao (2022), the resulting dataset spans 2019–2022 for the core panel used in the paper, with 2018–2023 used for broader descriptive context. The regression estimates are reported on the March 2020 to November 2022 window.

3.2 Data Contribution and Measurement Advantages

The data are a central contribution of this paper. Most early evidence on SMEs during COVID-19 relied on rapid-response surveys. Those surveys were valuable for tracking liquidity stress, expectations, and self-reported operating difficulties in real time, but they also came with familiar limitations: sample-based coverage, intermittent fieldwork, attrition, survivorship concerns, and weaker measurement of realized firm demography. The administrative register used here addresses those issues in several ways.

First, coverage is population-wide rather than sample-based, which substantially reduces selection concerns in the measurement of SME dynamics. Instead of observing a subset of firms willing and able to answer a survey during a crisis, we observe the near-universe of registered enterprises across Georgia’s 11 regions. Second, the monthly frequency allows us to track short-run changes in firm demography as conditions evolved, rather than relying on intermittent survey waves. This matters because the timing of lockdowns, reopenings, and later recovery was sharp, and entry responses in particular moved quickly.

Third, the outcomes are realized administrative events rather than perceptions or expectations. We observe registrations, closures, and operating stocks directly, which makes the data better suited to studying actual business turnover. Fourth, attrition is much less of a concern than in survey panels, where non-response and survival bias can become especially severe during crisis periods. Finally, the register can be linked directly to regional mortality and nightlight indicators at monthly frequency, which allows us to combine firm demography with local pandemic intensity and regional economic conditions in a way that most survey-based datasets cannot. Table 2 summarizes the main variables in the integrated panel.

TABLE 2 ABOUT HERE

3.3 Descriptive Statistics

For descriptive figures we exploit the full coverage of 2018–2023 to document longer-term trends in firm dynamics. For the regression analysis, however, we restrict attention to March 2020 through November 2022, the

common window over which the outcome variables, mortality series, lag structure, and controls are aligned in the final region-month estimating sample.

Table 3 reports summary statistics for the region-month variables used in the regression window and is most informative when read through the distribution of each variable, especially the percentiles, because the means can be pulled upward by large regions such as Tbilisi. For example, the mean number of monthly openings is 264.7, but that average masks substantial skewness driven by large regions; the median is much lower. Net firm creation is defined here as openings minus closings. The table highlights substantial variation across regions and months in both business dynamics and pandemic intensity. COVID-19 mortality per 100,000 population varies widely across region-month observations, and the satellite-based proxy for regional economic conditions also varies considerably across regions and months, underscoring the importance of controlling for local conditions in the panel regressions. Appendix Tables A1 and A2 complement these pooled moments with the full range of the distributions and with within-region dispersion, which is the variation most directly exploited by the fixed-effects models.

TABLE 3 ABOUT HERE

Beyond the overall summary, it is important to examine the composition of firms by size and industry. Table 4 reports the distribution of active enterprises across sectors and firm size categories during 2018–2023. Although the underlying register covers the full firm population, the analysis in this paper focuses on SMEs. Firm size categories follow the classification recorded in the Business Register.¹ Within the SME population, small enterprises account for 99.2% of firms, underscoring the central role of very small businesses in Georgia’s economy. Trade is the dominant sector, representing 42.2% of all SMEs, followed by services (17.1%) and manufacturing (7.4%). Medium-sized enterprises account for only 0.6% of the SME population and are most prevalent in services (22.7%) and trade (33.8%).

Additional descriptive evidence reinforces the later econometric emphasis on entry rates and firm size. Appendix Figure A1 and Appendix Table A3 show that the national small-firm openings rate falls from 14.2 per 1,000 lagged active firms in the 2019 baseline to 6.1 during the first lockdown, before recovering above baseline in 2022. Medium-firm entry also contracts during the lockdown, but remains more volatile and does not recover in the same way. Appendix Figure A2 further shows that the descriptive collapse in small-firm openings is concentrated in services and trade, with manufacturing and construction also weakening materially. Appendix Table A4 reports the mapping from NACE Rev. 2 industries to the broad sector groups used throughout the paper. These descriptive differences anticipate the regression evidence reported later, where the entry response is also strongest among small firms and varies with sector composition. The same figure shows an exceptional

¹Firm size categories follow the classification recorded in the Business Register, which Georgia adopted from the European Commission definition in 2017. Under this definition, small enterprises are those with up to 50 employees and annual revenues not exceeding 12 million GEL, while medium-sized enterprises have up to 249 employees and annual revenues not exceeding 60 million GEL; firms above these thresholds are classified as large. Size is assigned based on the firm’s characteristics at registration and updated in the Register when a firm’s reported revenues or employment cross a threshold, meaning firms can in principle move between categories over time.

spike in service-sector openings in March 2022. A plausible interpretation is that this reflects the coincidence of a major reopening of hospitality, events, and entertainment activities from March 2022 onward and a broader surge in demand, tourism, money transfers, immigration-related inflows, and new business formation after Russia’s invasion of Ukraine (Civil Georgia, 2022a,b; International Monetary Fund, 2022; World Bank, 2022; Transparency International Georgia, 2022).

TABLE 4 ABOUT HERE

Figure 1 presents monthly COVID-19 mortality by region. Regional heterogeneity is substantial: while Tbilisi recorded the highest absolute number of deaths, smaller regions such as Racha-Lechkhumi also experienced pronounced peaks in mortality during late 2021. Rural regions such as Racha-Lechkhumi and Imereti appear relatively more exposed, potentially reflecting differences in health infrastructure and local vulnerability.

FIGURE 1 ABOUT HERE

Company closures during 2019–2023 are concentrated entirely among small enterprises, underscoring their heightened vulnerability. Annual closures fell from 1,788 in 2019 to 1,325 in 2020, consistent with temporary government interventions and administrative delays during the first year of the pandemic. Closures then increased to 2,325 in 2021, 3,780 in 2022, and 4,666 in 2023, indicating that exit pressures intensified after the initial shock.

For brevity, additional descriptive figures on company openings and closings are reported in Appendix Figures A3 and A4. Together, these descriptive statistics underscore the dominant role of small firms in Georgia, the uneven regional impact of the pandemic, and substantial variation in firm openings, closings, and net creation across time and regions.

3.4 Stylized Facts Before Estimation

Before presenting the econometric analysis, three descriptive patterns motivate the empirical specification. First, openings respond quickly to crisis conditions: monthly openings fall sharply during pandemic peaks (Appendix Figure A3). Second, closures adjust with delay: annual closures are lower in 2020 but rise strongly afterward, indicating lagged exit dynamics. Third, exposure and structure are heterogeneous: pandemic mortality varies markedly across regions (Figure 1), and the enterprise base is heavily concentrated in small firms and a few sectors (Table 4).

These patterns motivate separate panel specifications for openings, closings, and net firm creation, with contemporaneous and lagged pandemic intensity and region and month-by-date fixed effects.

4 Methodology

4.1 Main Specification and Estimands

Our main econometric design is a region-month panel with region fixed effects and month-by-date fixed effects. For each outcome $k \in \{\text{Openings, Closings, Net Firm Creation}\}$, we define a rate per 1,000 lagged active SMEs and estimate:

$$R_{r,t}^k = \alpha^k + \sum_{j=0}^3 \beta_j^k D_{r,t-j} + \gamma^k N_{r,t} + \delta^k Pop_{r,t} + \theta^{k'} S_{r,t}^k + \mu_r^k + \mu_t^k + \varepsilon_{r,t}^k, \quad (1)$$

where $R_{r,t}^k$ is the monthly rate of openings, closings, or net firm creation per 1,000 lagged active SMEs in region r and month t ; $D_{r,t-j}$ is COVID-19 deaths per 100,000 population at lags $j = 0, \dots, 3$; $N_{r,t}$ is satellite-based nightlight intensity, used as a proxy for regional economic conditions; $Pop_{r,t}$ is regional population; and $S_{r,t}^k$ is an outcome-specific vector of sector-composition controls included in the main specification. Months with no recorded mortality in the estimation window are coded as zero before constructing lags.

For interpretation, β_0^k captures the contemporaneous within-region association, β_1^k to β_3^k capture delayed associations over the next three months, and $\sum_{j=0}^3 \beta_j^k$ is the four-month cumulative association. For example, if $\beta_0^k = -0.05$ in the openings equation, then an increase of 20 deaths per 100,000 population in a given region-month is associated with about 1 fewer firm opening for every 1,000 SMEs active in the previous month, holding fixed the controls and fixed effects.

4.2 Identification and Interpretation

The identifying variation comes from within-region changes in pandemic intensity over time, after absorbing time-invariant regional heterogeneity (μ_r) and nationwide month shocks (μ_t). Because major containment and support policies were often implemented nationally, the coefficients are interpreted as reduced-form associations between local pandemic pressure and firm-demography outcomes within a common policy environment, conditional on fixed effects and controls.

4.3 Inference with Few Clusters

Inference is based on robust standard errors clustered at the region level. Since the panel includes 11 regions, conventional clustered inference is fragile. We therefore complement clustered standard errors with wild-cluster bootstrap p -values for the mortality terms in the main specification and emphasize cumulative-effect tests alongside conventional significance levels.

4.4 Robustness Design

We assess specification sensitivity in two dimensions. First, our main dynamic specification includes outcome-specific sector composition controls, and we report a simpler comparison specification without sector shares to show how much they matter for precision and interpretation. Second, we retain level-based fixed-effects models as an additional comparison. Robustness is evaluated by whether signs, timing patterns, and cumulative associations remain stable across these alternatives.

5 Results

5.1 Main Results

Table 5 reports the main dynamic fixed-effects estimates on the common paper window from March 2020 to November 2022. The specification conditions on regional sector composition in addition to the standard controls and fixed effects. A simpler version without sector shares yields similar signs but weaker precision and is reported in Appendix Table A5.

In Table 5, the sector-controlled model gives a contemporaneous mortality coefficient of -0.056 for openings, with a conventional clustered p -value of 0.034 and a wild-cluster bootstrap p -value of 0.013. Interpreted at realistic changes in mortality, the magnitude is not trivial. A 20-deaths-per-100,000 increase in local mortality is associated with about 1.1 fewer openings for every 1,000 active SMEs in the previous month, which is a little over 8% of the sample mean openings rate. A 40-deaths-per-100,000 increase, close to the 90th percentile of monthly mortality in the regression sample, implies about 2.2 fewer openings per 1,000 active SMEs, or roughly 16% of the sample mean. The second lag is also negative and marginally informative, but the four-month cumulative effect remains imprecisely estimated. The estimate is very close in magnitude to the corresponding estimate in Appendix Table A5; the main difference is precision, not sign or scale. Conditioning on regional sector composition appears to remove some of the residual variation that otherwise blurs the entry response. Table 5 therefore sharpens an existing pattern rather than overturning the simpler comparison specification: pandemic intensity is most clearly related to lower SME entry rates on impact, not to a tightly estimated medium-run cumulative effect.

For net firm creation, the contemporaneous coefficient is also negative in the main specification (-0.034), but the small-cluster bootstrap weakens the inference substantially. The point estimate is therefore suggestive, but too imprecise to support a firm conclusion. For closings, neither the contemporaneous coefficient nor the lag structure supports a strong mortality effect. Across specifications, the closure margin remains the weakest part of the reduced-form evidence.

The administrative data clearly document pandemic-era disruption in SME dynamics, but the clearest reduced-form relationship with local mortality appears on the entry margin. This is broadly consistent with the

COVID–SME literature, which has repeatedly emphasized acute disruption among smaller firms and a sharp deterioration in operating conditions (Bartik et al., 2020; Cowling et al., 2020; Alfonsi et al., 2021; O’Toole et al., 2021; Wieczorek-Kosmala et al., 2021). What the present data add is that, once one looks at realized monthly firm demography rather than survey responses, the most visible adjustment in Georgia appears on business formation rather than on a tightly estimated contemporaneous closure spike. Evidence for broader effects on net firm growth and closures is weaker and should be treated cautiously.

TABLE 5 ABOUT HERE

5.2 Descriptive Regional and Sector Heterogeneity

Before turning to firm-size splits in the regression framework, it is useful to document how sharply the entry collapse differed across locations and sectors during the initial pandemic disruption. Table 6 compares average monthly *small-firm* openings in the 2019 baseline with the first lockdown period (March–May 2020). The purpose is descriptive rather than causal: it highlights where the entry slowdown was most pronounced before conditioning on fixed effects and controls.

The regional and sectoral differences are sizeable. Across regions, the proportional decline in small-firm entry ranges from about 22% in Racha-Lechkhumi and Kvemo Svaneti to roughly 60% in Kvemo Kartli, with large contractions also visible in Adjara, Tbilisi, Samegrelo-Zemo Svaneti, and Samtskhe-Javakheti. Across broad sectors, the steepest descriptive contractions appear in services, public activities, manufacturing, and trade, while construction and primary activities also contract noticeably. Utilities show the largest percentage decline, but that category starts from a very small baseline. These comparisons help motivate the emphasis on the entry margin in the next subsection, but they should not be interpreted as identifying sector-specific or region-specific treatment effects.

TABLE 6 ABOUT HERE

5.3 Firm Size Heterogeneity

Table 7 asks whether the entry result is shared uniformly across SMEs or concentrated among smaller firms. The answer is clear: the pooled openings-rate result is driven by *small* firms rather than *medium* firms.

In the sector-controlled specification, the contemporaneous mortality coefficient for small-firm openings is -0.055 , with a clustered p -value of 0.036 and a wild-cluster bootstrap p -value of 0.022. Put differently, a 10-deaths-per-100,000 increase in local mortality is associated with roughly 0.55 fewer small-firm openings per 1,000 lagged active small firms in the same month, or about 4% of the mean small-firm openings rate in the estimation sample. By contrast, the corresponding coefficient for medium-firm openings is -0.060 , similar in sign and scale but far more imprecisely estimated, with a bootstrap p -value of 0.353. The four-month cumulative effects are insignificant for both groups.

The heterogeneity result sharpens the interpretation of the paper. Pandemic intensity appears to have reduced entry mainly among smaller firms within the SME distribution, while medium-firm entry was more resilient. This is economically plausible given tighter financing constraints, smaller cash buffers, and greater sensitivity to uncertainty among smaller firms.

We also estimated analogous size-split specifications for net firm creation and closings and report them in Appendix Table A6. These extensions do not provide a cleaner or more informative result than the entry regressions. Small-firm net-firm-creation coefficients remain negative but lose precision under wild-cluster bootstrap inference, while the medium-firm net-firm-creation result is harder to interpret because medium-firm closings are essentially absent in the estimation window. The closure margin itself remains weak throughout. For that reason, the most informative heterogeneity result remains the effect on small-firm entry.

TABLE 7 ABOUT HERE

5.4 Robustness Checks

Table 8 brings together two targeted robustness checks and two falsification exercises for the small-firm entry result. The first robustness check replaces the monthly openings-rate outcome with a year-over-year difference, which is a stricter way to absorb seasonality by comparing each month to its counterpart one year earlier. The second augments the main specification with region-specific linear trends, allowing for gradual differential trajectories across regions. The placebo exercises then ask whether the result survives simple timing tests. Adding leads of the mortality variable to the small-firm entry specification does not suggest anticipatory effects: the lead coefficients are individually insignificant and jointly insignificant. Assigning the later mortality path to the pre-pandemic period as a shifted placebo also produces no meaningful association with small-firm entry. Full placebo estimates are reported in Appendix Tables A7 and A8.

The contemporaneous mortality coefficient remains negative in both of the tougher non-placebo checks. In the year-over-year specification the estimate becomes somewhat larger in magnitude (-0.071) but less precise, which is consistent with the lower signal-to-noise ratio of differenced outcomes. In the region-trend specification the coefficient stays very close to the main estimate (-0.054 versus -0.055 in Table 7) and remains marginally informative, which is reassuring given the concern that some regions may have been on gradually different trajectories even without the pandemic. The placebo exercises point in the same direction. Once future mortality leads are added, those lead coefficients are individually insignificant and jointly insignificant, so the timing pattern does not suggest that later deaths are spuriously predicting current entry. Likewise, assigning the later mortality path to the pre-pandemic period produces a coefficient close to zero and no meaningful cumulative effect. These checks make the main small-firm entry result more credible: the sign survives tougher specifications, while the falsification exercises yield null results.

TABLE 8 ABOUT HERE

The robustness exercises point to two conclusions. First, the negative association between mortality and SME openings is more stable than the corresponding associations for net firm creation and closings. It appears in both pooled tables, remains statistically informative in the sector-controlled specification under wild-cluster bootstrap inference, is shown in Table 7 to be concentrated in small rather than medium firms, and survives the additional specification checks in Table 8. Second, the broader firm-demography effects are not comparably robust: cumulative effects are imprecise, the net-firm-creation result weakens under bootstrap inference, and the closure margin remains close to zero throughout.

The level-based comparison models, reported in Appendix Table A5, are weaker than the main specification and do not overturn this interpretation. Local pandemic intensity is associated mainly with weaker entry dynamics, especially among small firms, while the evidence for broader effects on net firm growth and firm exit remains tentative.

6 A simple model

This section presents a simple model that rationalizes two features of firm dynamics during COVID-19. First, fewer firms enter when COVID-19 prevalence is higher, and the firms deterred from entering would have operated at a smaller scale. At the same time, among firms already operating during the pandemic, a subsequent intensification of COVID-19 need not lead to exit.

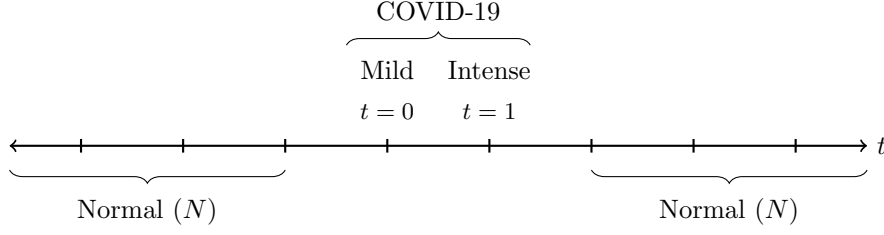
For potential entrants, a worsening of COVID-19 reduces entry because the expected return to production is lower. This can be rationalized by weaker demand, tighter activity restrictions, or other disruptions that reduce the payoff from operating during the pandemic. For incumbent firms, the mechanism operates through selection. Firms that remain active during COVID-19 are positively selected, either because entry costs and fixed operating costs are high in normal times or because the initial COVID-19 shock induces the exit of low-productivity firms. As a result, a subsequent increase in the severity of COVID-19 need not lead incumbent firms to exit.

6.1 Aggregate states and timing

We model changes in demand conditions through an aggregate TFP shifter that scales firm revenue. This is a reduced-form representation of demand rather than a literal change in production technology. Holding inputs and firm productivity fixed, a decline in TFP lowers the revenue generated by a given production plan. It can therefore represent fewer units being sold because of weaker demand, a decline in the prices firms receive, or a combination of both. Before COVID-19, aggregate TFP is normalized to one. COVID-19 lasts for two periods. Period 0 is the unexpected outbreak, with TFP A_0 , and period 1 is the more severe COVID-19 period, with TFP A_1 :

$$1 > A_0 > A_1 > 0. \quad (2)$$

After period 1, TFP returns permanently to one.



Incumbent firms operate with installed capital. For these firms, capital is chosen in advance and cannot be adjusted instantaneously at the time of production. This creates a planning lag: when an incumbent enters period t , its productive capacity is partly predetermined by the capital installed in the previous period. Thus, incumbent firms carry legacy capital into the period, while new entrants can choose their initial scale at the time of entry.

At the beginning of each period, a mass of potential firms draws permanent productivity z from a common exogenous distribution and decides whether to enter. An entrant pays a sunk entry cost $F > 0$, selects its initial capital K_t , and produces immediately. From the next period onward, it faces the same planning lag as every other incumbent. A firm may instead exit, in which case its value is normalized to zero.

In normal times, aggregate conditions are constant, so an incumbent chooses the same capital stock in consecutive periods. Thus, although capital is selected one period in advance, no capital adjustment occurs along the normal path.

The entry cost can be interpreted as including the organizational, search, and installation expenditures required to create a new operation. Under this interpretation, paying F allows a new firm to install capital adapted to current conditions without first reorganizing a legacy capital stock. This distinction between new investment and the reorganization of installed capital applies in every aggregate state and is not specific to COVID-19.^{2,3}

6.2 Technology and static operating decisions

A firm with productivity $z > 0$, capital $K \geq 0$, and employment $L \geq 0$ produces revenue

$$A_s z K^\alpha L^\gamma, \quad \alpha > 0, \quad \gamma > 0, \quad \alpha + \gamma < 1, \quad (3)$$

²An alternative would impose the same one-period planning lag on potential entrants. Firms observed entering in period 0 would then have made their entry and capital decisions before the unexpected outbreak, under normal-time expectations. Entry in period 1, chosen after the outbreak, would respond to the lower expected profitability of the severe period. This alternative timing would make the decline in entry even easier to generate, but part of the result would be mechanical because period-0 entry was already predetermined.

³An otherwise identical model without planning lags could also generate the empirical patterns we document. For no firm to exit when COVID-19 becomes more severe, pre-pandemic selection into incumbency would have to be sufficiently strong. However, this would imply that no incumbent firm exits during the initial COVID-19 period.

where $s \in \{N, 0, 1\}$, and N denotes normal times. The wage is $w > 0$, the fixed operating cost is $c > 0$, and rK is the per-period cost of using capital in production. Conditional on installed capital, the firm chooses employment to solve

$$\pi_s(z, K) = \max_{L \geq 0} \{A_s z K^\alpha L^\gamma - wL - rK - c\}. \quad (4)$$

The optimal employment choice is

$$L_s^*(z, K) = \left(\frac{\gamma A_s z K^\alpha}{w} \right)^{\frac{1}{1-\gamma}}. \quad (5)$$

Substitution gives

$$\pi_s(z, K) = \kappa_L (A_s z K^\alpha)^{\frac{1}{1-\gamma}} - rK - c, \quad (6)$$

where

$$\kappa_L = (1 - \gamma) \gamma^{\frac{\gamma}{1-\gamma}} w^{-\frac{\gamma}{1-\gamma}}. \quad (7)$$

When capital can be selected for the current period, define

$$\pi_s^*(z) = \max_{K \geq 0} \pi_s(z, K). \quad (8)$$

Let

$$\sigma = \frac{1}{1 - \alpha - \gamma} \quad (9)$$

and

$$\Omega = (1 - \alpha - \gamma) (\alpha^\alpha \gamma^\gamma r^{-\alpha} w^{-\gamma})^\sigma. \quad (10)$$

The static solution is

$$Y_s^*(z) = (A_s z \alpha^\alpha \gamma^\gamma r^{-\alpha} w^{-\gamma})^\sigma, \quad (11)$$

$$K_s^*(z) = \frac{\alpha}{r} Y_s^*(z), \quad (12)$$

$$L_s^*(z) = \frac{\gamma}{w} Y_s^*(z), \quad (13)$$

$$\pi_s^*(z) = \Omega (A_s z)^\sigma - c. \quad (14)$$

Thus productivity, capital, and employment move together. Firms with lower productivity optimally operate with fewer workers.

6.3 The normal economy

Firms discount future profits by $\beta \in (0, 1)$. In the permanent normal state, aggregate TFP is $A_N = 1$, and an incumbent has value

$$V_N(z) = \frac{\Omega z^\sigma - c}{1 - \beta}. \quad (15)$$

A potential entrant pays F and enters if $V_N(z) \geq F$. The normal entry cutoff is

$$z_E^N = \left[\frac{c + (1 - \beta)F}{\Omega} \right]^{1/\sigma}. \quad (16)$$

Normal incumbents therefore satisfy $z \geq z_E^N$ and enter the outbreak with capital $K_N^*(z)$.

6.4 Period 1

During period 0, a surviving incumbent knows that TFP in period 1 will be A_1 . It therefore selects $K_1^*(z)$ for the following period. At the end of period 1, it selects normal capital for the permanent return to the normal state. Hence, its continuation value after period 1 is $V_N(z)$. The gross value of operating in period 1 is

$$W_1(z) = \pi_1^*(z) + \beta V_N(z) = \frac{\Omega [(1 - \beta)A_1^\sigma + \beta] z^\sigma - c}{1 - \beta}. \quad (17)$$

An incumbent continues if $W_1(z) \geq 0$.

A potential entrant in period 1 can install $K_1^*(z)$ immediately. Its net entry value is

$$J_1(z) = W_1(z) - F. \quad (18)$$

The period-1 entry cutoff is therefore

$$z_E^1 = \left[\frac{c + (1 - \beta)F}{\Omega [(1 - \beta)A_1^\sigma + \beta]} \right]^{1/\sigma}. \quad (19)$$

6.5 Entry in period 0

A period-0 entrant selects $K_0^*(z)$, produces immediately, and then selects $K_1^*(z)$ for period 1. Its net value is

$$J_0(z) = \pi_0^*(z) + \beta \max\{0, W_1(z)\} - F. \quad (20)$$

The period-0 entry cutoff satisfies $J_0(z_E^0) = 0$.

Proposition 1 (Entry). Suppose

$$A_0^{\frac{1}{1-\alpha-\gamma}} > (1-\beta)A_1^{\frac{1}{1-\alpha-\gamma}} + \beta. \quad (\text{E})$$

Then $z_E^1 > z_E^0$, so fewer firms enter in period 1 than in period 0. Firms deterred from entering in period 1 would have operated with lower employment than firms that enter in period 1. The proof is reported in [Appendix B](#).

There are two opposing forces shaping entrants' decisions in periods 0 and 1. TFP is lower in period 1 than in period 0. However, firms know that COVID-19 will disappear after period 1. Condition (E) states that, for entry to be lower in period 1 than in period 0, TFP in period 1 must be sufficiently low and/or firms must discount the future sufficiently strongly.

6.6 Incumbent exit

An incumbent enters period 0 with normal capital $K_N^*(z)$. After reoptimizing employment but before changing its capital for the following period, its current operating profit is derived as follows. Let $Y_N^*(z)$ denote optimal output in normal times. The normal-state first-order conditions and the definition of Ω imply

$$rK_N^*(z) = \alpha Y_N^*(z), \quad (21)$$

$$\Omega z^\sigma = (1 - \alpha - \gamma) Y_N^*(z). \quad (22)$$

Holding $K_N^*(z)$ fixed and reoptimizing employment after TFP falls to A_0 gives

$$Y_0^I(z) = A_0^{\frac{1}{1-\gamma}} Y_N^*(z). \quad (23)$$

Since optimal employment satisfies $wL_0^I(z) = \gamma Y_0^I(z)$, it follows that

$$\pi_0^I(z) = (1 - \gamma) Y_0^I(z) - rK_N^*(z) - c = \left[(1 - \gamma) A_0^{\frac{1}{1-\gamma}} - \alpha \right] Y_N^*(z) - c. \quad (24)$$

Using equation (22) to substitute for $Y_N^*(z)$, the incumbent's profit can be written entirely as a function of productivity:

$$\pi_0^I(z) = \frac{(1 - \gamma) A_0^{\frac{1}{1-\gamma}} - \alpha}{1 - \alpha - \gamma} \Omega z^\sigma - c. \quad (25)$$

The difference between $\pi_0^I(z)$ and $\pi_0^*(z)$ captures the burden of legacy capital. The former evaluates period-0 profitability at capital chosen for the normal state, whereas the latter allows capital to be selected for current conditions.

An incumbent's value at the outbreak is

$$W_0^I(z) = \pi_0^I(z) + \beta \max\{0, W_1(z)\}. \quad (26)$$

It exits when $W_0^I(z) < 0$. If it continues, it chooses $K_1^*(z)$ and enters period 1 with capital adapted to the severe COVID-19 state.

Proposition 2 (Incumbent exit). Suppose

$$\frac{(1-\gamma)A_0^{\frac{1}{1-\gamma}} - \alpha}{1-\alpha-\gamma} < (1-\beta)A_1^{\frac{1}{1-\alpha-\gamma}} + \beta. \quad (S)$$

Then no incumbent operating in period 0 exits when the economy moves to period 1. The proof is reported in [Appendix B](#).

As in Proposition 1, there are also two opposing forces shaping incumbents' exit decisions between periods 0 and 1. TFP is lower in period 1, reducing current operating profits. However, firms enter period 1 with capital chosen for that period and know that normal conditions will return afterward. Condition (S) states that the profitability associated with adapted capital and the anticipated return to normal must be sufficiently high relative to period-0 profitability with legacy capital. Equivalently, TFP in period 1 must not be too low and/or firms must place sufficient weight on the future. Under this condition, every incumbent willing to operate in period 0 has a positive continuation value in period 1.

6.7 Interpretation

The central parameter ordering is

$$\frac{(1-\gamma)A_0^{\frac{1}{1-\gamma}} - \alpha}{1-\alpha-\gamma} < (1-\beta)A_1^{\frac{1}{1-\alpha-\gamma}} + \beta < A_0^{\frac{1}{1-\alpha-\gamma}}. \quad (27)$$

The first term measures the profitability of an incumbent in period 0 with capital inherited from normal times. The last term measures the profitability of a new entrant that selects capital for period 0. The middle term combines optimized profitability in period 1 with the subsequent return to the normal state.

The first inequality ensures that a firm willing to operate with legacy capital in period 0 has positive continuation value once its capital has been adapted to period 1. The second inequality makes entry less attractive in period 1 than in period 0.

6.8 Initial exit at the unexpected outbreak

The absence of incumbent exit between periods 0 and 1 does not imply that no incumbents exit when the economy moves from the pre-COVID normal state to period 0. In particular, some normal incumbents exit at the unexpected outbreak if

$$0 < (1 - \beta) \frac{(1 - \gamma)A_0^{\frac{1}{1-\gamma}} - \alpha}{1 - \alpha - \gamma} + \beta \left[(1 - \beta)A_1^{\frac{1}{1-\alpha-\gamma}} + \beta \right] < \frac{c}{c + (1 - \beta)F}. \quad (I)$$

To see this, rescale productivity as $x = \Omega z^\sigma$. The least productive normal incumbent has $x_E^N = c + (1 - \beta)F$. When its period-1 value is positive, its outbreak value satisfies

$$(1 - \beta)W_0^I(x_E^N) = \left\{ (1 - \beta) \frac{(1 - \gamma)A_0^{\frac{1}{1-\gamma}} - \alpha}{1 - \alpha - \gamma} + \beta \left[(1 - \beta)A_1^{\frac{1}{1-\alpha-\gamma}} + \beta \right] \right\} x_E^N - c < 0. \quad (28)$$

If its period-1 value is nonpositive, condition (S) implies the same conclusion from its current operating profit alone. By continuity, an interval of the least productive normal incumbents exits. The lower inequality in (I) also ensures that sufficiently productive incumbents survive.⁴

7 Conclusion

This paper uses population-level administrative business-register data from Georgia to provide new evidence on how firm dynamics adjusted during the COVID-19 pandemic. While much of the existing literature has relied on surveys to document business distress and expected outcomes, our data allow us to track realized firm openings, closings, and net firm creation at monthly frequency for the near-universe of registered SMEs.

The data show substantial disruption in SME activity during the pandemic, especially on the entry margin. Higher mortality is associated with lower SME opening rates, particularly once regional sector composition is taken into account. Interpreted at realistic changes in mortality, the magnitude is not negligible: in the small-firm specification, a 20-deaths-per-100,000 increase in local mortality is associated with about 1.1 fewer openings per 1,000 active small firms in the previous month, or a little over 8% of the average small-firm openings rate in the estimation sample. A 40-deaths-per-100,000 increase, close to the 90th percentile of monthly mortality in the regression sample, implies about 2.2 fewer openings per 1,000 active small firms, or roughly 16% of the average small-firm openings rate. The size split suggests that this relationship is concentrated among small

⁴For example, all three conditions (E), (S), and (I) hold when $\alpha = 0.3$, $\gamma = 0.5$, $\beta = 0.9$, $A_0 = 0.99$, $A_1 = 0.872$, $c = 1$, $F = 0.3125$, and $r = w = 1$. Under these values, the three terms in the central ordering are $0.95025 < 0.9504176 < 0.9509900$, while the middle expression in (I) is $0.9504009 < 0.9696970 = c/[c + (1 - \beta)F]$.

firms, while medium-firm entry appears more resilient. The additional year-over-year and region-trend checks preserve the negative sign of the small-firm entry result, and the placebo exercises do not indicate spurious timing. By contrast, the evidence for net firm creation is weaker under small-cluster inference, and the closure margin remains imprecisely estimated.

These patterns also underscore the value of administrative business-demography data for studying crisis adjustment. Such data make it possible to move beyond self-reported distress and observe realized changes in business activity across the main margins of firm dynamics. In the Georgian case, the clearest response is on the entry margin: business formation declines when local mortality rises, while the evidence on closures is more limited and less precisely estimated.

The simple model helps interpret this asymmetry. When pandemic conditions worsen, prospective entrants can wait, while incumbents have already incurred the sunk cost of entry and continue operating with installed capacity. In that setting, a negative shock is more likely to appear first as missing entry, especially among smaller firms, than as an immediate increase in formal closures. The Georgian evidence is consistent with that logic.

The policy implications concern the entry margin. If pandemic disruption falls most heavily on small-firm entry, then crisis policy should not focus only on preserving incumbents. It should also pay attention to the conditions under which new small businesses are formed, particularly where tighter financing constraints and thinner buffers make entrepreneurial activity more fragile. The results do not point to large aggregate effects on all margins of firm demography, but they do suggest that support aimed at prospective small entrants may matter for business formation during large shocks.

Future work should link these data to richer measures of policy stringency, financial support, and sector-specific exposure. More broadly, the paper suggests that during large adverse shocks, preserving incumbent firms is not the same as preserving the flow of new firm creation.

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Table 1: Timeline of COVID-19 policy measures in Georgia, 2020

Date	Measure
2020-01-29	Air traffic to China suspended.
2020-02-23	Air traffic with Iran restricted.
2020-02-26	First COVID-19 case confirmed.
2020-03-01	Educational process in schools, kindergartens, and universities stopped.
2020-03-06	9 cases confirmed.
2020-03-12	25 cases confirmed.
2020-03-14	30 cases confirmed.
2020-03-16	Mountain and ski resorts temporarily closed.
2020-03-16	Cafés, bars, and restaurants limited to delivery service only.
2020-03-18	Movement of minibus taxis prohibited.
2020-03-18	Borders of Georgia closed.
2020-03-19	Shopping facilities closed (exceptions: groceries, pharmacies, gas stations, post offices, banks).
2020-03-21	State of emergency declared.
2020-03-23	All intercity transport restricted.
2020-03-23	All facilities closed except pharmacies and grocery stores.
2020-03-23	Entry/exit restricted for Marneuli and Bolnisi municipalities.
2020-03-29	Samegrelo–Zemo Svaneti: measures tightened; public transport banned in region; checkpoints at municipal entries/exits.
2020-03-31	Curfew 01:00–06:00 announced.
2020-03-31	All public transport (incl. metro) suspended; only light vehicles allowed (max 3 persons per vehicle).
2020-03-31	Checkpoints organized in Tbilisi, Batumi, Kutaisi, Rustavi, Poti, Zugdidi, and Gori.
2020-04-01	For 3 months, utilities fully financed for consumers under 200 kWh electricity and 200 m ³ gas.
2020-04-07	Agricultural markets closed.
2020-04-10	Full quarantine announced in Lentekhi region.
2020-04-12	Full quarantine in Kobuleti municipality units (Gvara, Mukhaestate, Leghva, Tskavroka); entry/exit banned.
2020-04-15	Entry/exit banned in Tbilisi, Rustavi, Kutaisi, and Batumi.
2020-04-17	Movement of light vehicles fully prohibited nationwide until April 27 (initially April 21).
2020-05-05	Permitted to operate: all construction/repair works; building materials production; auto service and car washes.
2020-05-05	Entry/exit restrictions lifted in Kutaisi and Batumi.

Date	Measure
2020-05-10	64 of 90 registered agricultural markets reopened.
2020-05-11	All manufacturing reopened; retail/wholesale shops with street entrance reopened (clothes/shoes shops excluded); malls remain closed.
2020-05-11	Movement restrictions in Tbilisi lifted.
2020-05-14	Entry/exit restrictions lifted in Rustavi and Gardabani.
2020-05-18	Restrictions lifted in Marneuli region.
2020-05-18	Beauty salons and aesthetic medicine centers reopened.
2020-05-22	Bolnisi municipality: measures tightened; village of Geta added to strict quarantine (alongside Mushevan).
2020-05-23	State of emergency ended; curfew canceled.
2020-05-29	All municipal transport, including metro, resumed.
2020-06-01	Opened: all shops (incl. malls); all markets (open/closed); restaurants with open-air seating.
2020-06-08	Intercity transport restored; hotels reopened subject to health permits.
2020-06-15	Domestic tourism reopened.
2020-06-15	Georgian Railways resumed passenger services.
2020-06-15	Strict quarantine lifted in Geta (Bolnisi).
2020-06-19	Allowed: outdoor seminars/trainings; adventure activities (rafting, kayaking, canyoning, etc.).
2020-07-01	Quarantine canceled in Mushevani (Bolnisi).
2020-07-01	EU opened air borders to Georgia.
2020-07-06	Allowed: trainings/conferences in closed spaces; outdoor attractions; vocational institutions; swimming pools and gyms (with regulations).
2020-07-13	Rehearsals permitted in closed spaces; open-air cultural events allowed (max 200 persons).
2020-07-20	Social/festive events allowed in open space (max 100 persons).
2020-07-24	Karajalar village (Gardabani) placed under quarantine.
2020-08-10	Quarantine canceled in Karajalar.
2020-08-10	Quarantine imposed in Mestia municipality (incl. Ushguli and Langer); entry/exit banned; movement between villages restricted.
2020-08-21	Quarantine restrictions lifted across Mestia municipality except Mestia township and Langer (remain under quarantine).
2020-09-02	Hotel “Oasis” in Chakvi closed.
2020-09-02	Quarantine lifted in Svaneti: Mestia township and Langer reopened.
2020-09-10	Checkpoints organized in Tsageri municipality villages Orbel and Lajana.
2020-09-11	Schools and universities in Tbilisi and other large cities start the term remotely.
2020-09-24	Classroom teaching resumes for grades I–VI in Tbilisi, Rustavi, Kutaisi, and Zugdidi; all grades resume in Poti and Gori from Oct 1; schools in Adjara continue online.

Date	Measure
2020-09-24	Adjara: restrictions tightened; public transport banned; restaurants limited to 22:00 closing.
2020-10-19	Higher and vocational education resumes in-person nationwide except in Adjara.
2020-11-09	Large cities: movement banned 22:00–05:00.
2020-11-28	From 21:00–05:00: movement on foot/vehicle and presence in public space prohibited; long-distance passenger transport (rail/bus/minibus) suspended; light vehicles (incl. taxis) exempt; restaurants switch to delivery only; fitness centers and pools closed; activities of sports, artistic, and cultural circles/studios restricted.

Table 2: Variables in the Integrated Dataset

Dimension	Variable	Unit / Coverage	Source
Enterprise-level	Openings	Monthly, by region	Business Register
	Closings	Monthly, by region	
	Operational status	Active; inactive; suspended	
	Enterprise age	Years since registration	
	Size classification	Micro; small; medium	
	Industry classification	NACE Rev. 2, 8 sectors	
	Geographic location	11 regions	
Health & demographics	COVID-19 deaths	Monthly, by region	NatStat Georgia
	COVID-19 death rate	Per 100,000 population	
	Population	By region and period	
	Mortality rate	Regional annual average	
Economic activity proxy	Nightlight intensity	Monthly, by region	IEG database

Table 3: Descriptive Statistics, Regression Window. Summary statistics are calculated on the March 2020 to November 2022 region-month regression sample ($N = 363$ region-month observations). Population is reported in thousands of persons for readability.

Variable	Mean	Std. Dev.	P10	P25	Median	P75	P90
SME openings (monthly, region)	264.68	386.34	35.20	67.00	145.00	272.50	579.20
SME closings (monthly, region)	19.02	22.02	1.00	5.00	12.00	26.00	39.80
SME net firm creation (monthly, region)	245.66	367.63	33.20	59.00	129.00	242.00	550.40
COVID-19 deaths (per 100,000)	13.57	19.00	0.00	0.00	4.63	22.43	39.00
Population (thousands, region)	337.43	304.93	92.35	108.10	304.92	434.24	486.98
Nightlight intensity (index)	18501.51	14543.96	0.00	0.00	20798.30	29768.59	35370.99

Table 4: Distribution of Companies by Size and Industry. Numbers in parentheses are row percentages. Data include registered and active enterprises, 2018–2023.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Size	Construction	Manufacturing	Primary	Public	Services	Trade	Other	Utilities	<i>Total</i>
Small	23,887 (6.56%)	26,707 (7.33%)	6,198 (1.70%)	17,904 (4.92%)	62,029 (17.04%)	153,823 (42.25%)	73,066 (20.07%)	491 (0.13%)	364,105 (100.00%)
Medium	296 (13.07%)	285 (12.59%)	42 (1.86%)	314 (13.87%)	514 (22.70%)	766 (33.83%)	10 (0.44%)	37 (1.63%)	2,264 (100.00%)
Large	31 (5.08%)	79 (12.95%)	14 (2.30%)	121 (19.84%)	113 (18.52%)	226 (37.05%)	0 (0.00%)	26 (4.26%)	610 (100.00%)
Total	24,214 (6.60%)	27,071 (7.38%)	6,254 (1.70%)	18,339 (5.00%)	62,656 (17.07%)	154,815 (42.19%)	73,076 (19.91%)	554 (0.15%)	366,979 (100.00%)

Table 5: COVID-19 and SME Dynamics: Dynamic Rate Estimates with Sector Composition

	(1) Openings rate	(2) Net firm creation rate	(3) Closings rate
Deaths (per 100k) $D_{r,t}$	-0.056** (0.026)	-0.034** (0.016)	-0.003 (0.003)
Deaths, lag 1 $D_{r,t-1}$	-0.002 (0.021)	-0.005 (0.017)	-0.001 (0.002)
Deaths, lag 2 $D_{r,t-2}$	-0.025* (0.015)	-0.006 (0.017)	-0.002 (0.002)
Deaths, lag 3 $D_{r,t-3}$	0.021 (0.034)	0.005 (0.025)	0.000 (0.003)
Four-month cumulative effect	-0.062 (0.074) [$p=0.407$]	-0.040 (0.038) [$p=0.298$]	-0.006 (0.004) [$p=0.135$]
Wild-bootstrap p for $D_{r,t}$	[0.013]	[0.128]	[0.570]
Nightlight intensity	0.00009* (0.00005)	0.00008** (0.00003)	-0.00000 (0.00000)
Population	0.00018 (0.00012)	0.00023* (0.00013)	-0.00000 (0.00000)
Observations	363	363	363
Outcome-specific sector shares	Yes	Yes	Yes
Region FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
SEs clustered by region	Yes	Yes	Yes

Notes: Fixed-effects regressions with region and month-by-date fixed effects on the March 2020 to November 2022 sample. Dependent variables are rates per 1,000 lagged active SMEs. Robust standard errors clustered at the region level are in parentheses. Brackets report p -values for four-month cumulative-effect tests and wild-cluster bootstrap p -values for the contemporaneous mortality term. Deaths variables are measured per 100,000 population. Sector-share controls are expressed in percentage points and vary by outcome definition: opening shares for the openings regression, operational shares for the net-firm-creation regression, and closing shares for the closings regression. Those sector-share coefficients are included in estimation but omitted from the table for readability. Inference is interpreted cautiously given 11 region clusters.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Descriptive Heterogeneity in Small-Firm Openings

Panel A. Regional Small-Firm Openings			
Region	2019 Avg.	Mar–May 2020 Avg.	% Change
Kvemo Kartli	192.42	76.67	-60.16
Autonomous Republic of Adjara	266.33	110.00	-58.70
Tbilisi	1048.67	497.33	-52.57
Samegrelo-Zemo Svaneti	170.42	84.00	-50.71
Samtskhe-Javakheti	64.83	32.00	-50.64
Mtskheta-Mtianeti	44.58	22.67	-49.16
Kakheti	136.33	69.67	-48.90
Imereti	282.50	152.67	-45.96
Shida Kartli	113.50	63.67	-43.91
Guria	56.42	32.00	-43.28
Racha-Lechkhumi and Kvemo Svaneti	14.92	11.67	-21.79
Panel B. Sectoral Small-Firm Openings			
Sector	2019 Avg.	Mar–May 2020 Avg.	% Change
Utilities	0.39	0.06	-84.62
Services	44.24	18.06	-59.18
Public	11.35	4.64	-59.15
Manufacturing	18.57	9.03	-51.37
Trade	96.18	48.73	-49.34
Construction	23.16	14.15	-38.89
Primary	5.08	3.24	-36.12

Notes: Panel A reports all 11 regions, and Panel B reports all broad sectors, comparing average monthly small-firm openings in the 2019 baseline with the first lockdown period (March–May 2020). These are descriptive comparisons included to summarize heterogeneity in the raw data rather than causal effects. Utilities should be interpreted cautiously because the underlying baseline level is very small.

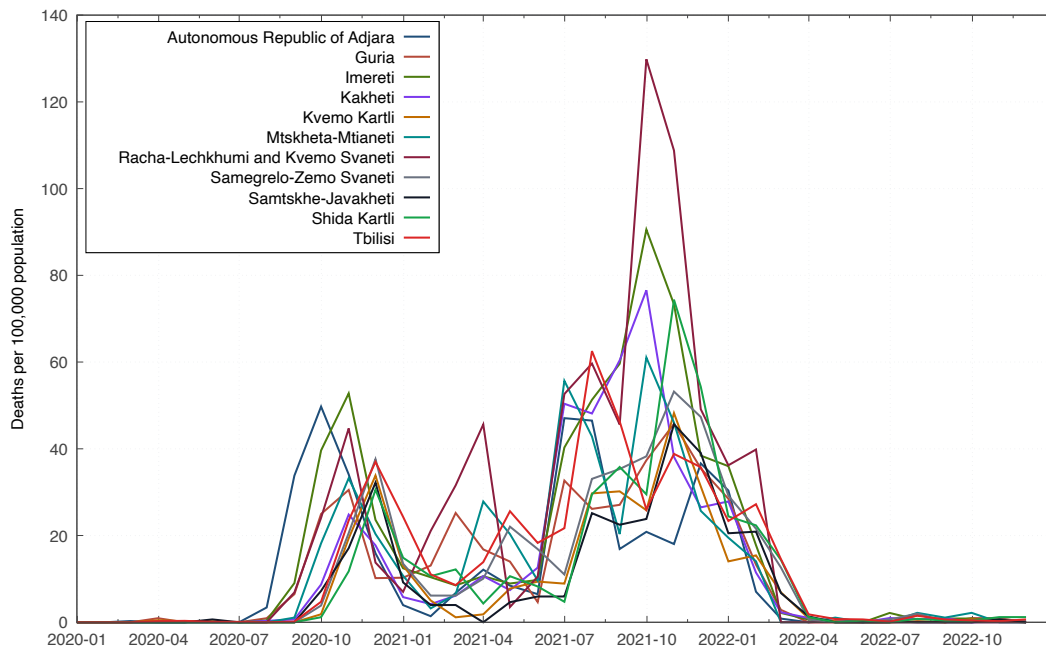
Figure 1: COVID-19 Deaths by Region (per 100,000 Population). Monthly distribution of COVID-19 deaths per 100,000 population across Georgian regions from 2020 to 2022. Source: National Statistics Office of Georgia.

Table 7: COVID-19 and SME Openings: Firm Size Heterogeneity

	(1) Small firms	(2) Medium firms
Deaths (per 100k) $D_{r,t}$	-0.055** (0.026)	-0.060 (0.054)
Deaths, lag 1 $D_{r,t-1}$	-0.004 (0.021)	0.044* (0.024)
Deaths, lag 2 $D_{r,t-2}$	-0.025 (0.016)	-0.032 (0.029)
Deaths, lag 3 $D_{r,t-3}$	0.024 (0.033)	0.019 (0.012)
Four-month cumulative effect	-0.061 (0.073)	-0.030 (0.051)
Wild-bootstrap p for $D_{r,t}$	$[p=0.407]$ [0.022]	$[p=0.564]$ [0.353]
Observations	363	363
Nightlights control	Yes	Yes
Population control	Yes	Yes
Size-specific sector shares	Yes	Yes
Region FE	Yes	Yes
Month FE	Yes	Yes
SEs clustered by region	Yes	Yes

Notes: Fixed-effects regressions with region and month-by-date fixed effects on the March 2020 to November 2022 sample. Dependent variables are openings rates per 1,000 lagged active firms within each size class. Robust standard errors clustered at the region level are in parentheses. Brackets report p -values for four-month cumulative-effect tests and wild-cluster bootstrap p -values for the contemporaneous mortality term. Sector-share controls are constructed separately for small and medium openings. Inference is interpreted cautiously given 11 region clusters.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Robustness of the Small-Firm Entry Result

	Main	Year-over-year	Region trends	Placebo leads	Shifted placebo
Main mortality term	-0.055** (0.026)	-0.071 (0.044)	-0.054* (0.028)	-0.025** (0.012)	0.003 (0.030)
Clustered p -value	0.036	0.107	0.055	0.041	0.930
Wild-bootstrap p -value	0.022	0.178	0.016	—	0.931
Four-month cumulative effect	-0.061	-0.075	-0.090	—	2.013
Cumulative-effect p -value	0.407	0.381	0.350	—	0.211
Placebo leads joint p -value	—	—	—	0.157	—
Observations	363	363	363	363	99
Month FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
Region-specific trends	No	No	Yes	No	No
Small-firm sector shares	Yes	Yes	Yes	Yes	Yes

Notes: The first three columns use the March 2020 to November 2022 region-month sample and measure mortality in deaths per 100,000 population. The main column reproduces the main small-firm openings-rate specification from Table 7. The year-over-year column uses the monthly change in the small-firm openings rate relative to the same calendar month one year earlier. The region-trend column adds region-specific linear time trends to the main specification. The placebo-leads column adds future mortality leads ($t+1$ to $t+3$) to the main specification; the reported joint p -value is a Wald test that those lead coefficients are jointly zero. The shifted-placebo column estimates the same small-firm openings-rate specification on the pre-COVID window from March to November 2019, assigning the later mortality path to that earlier period as a fake treatment. Robust standard errors are clustered at the region level. Wild-bootstrap p -values are shown where directly relevant to the placebo design.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

A Further tables and figures

Table A1: Extended Descriptive Statistics, Regression Window. Extended pooled summary statistics for the March 2020 to November 2022 region-month regression sample ($N = 363$). Population is reported in thousands of persons.

Variable	Mean	Std. Dev.	P10	P25	Median	P75	P90	Min	Max
SME openings (monthly, region)	264.68	386.34	35.20	67.00	145.00	272.50	579.20	1.00	2394.00
SME closings (monthly, region)	19.02	22.02	1.00	5.00	12.00	26.00	39.80	0.00	138.00
SME net firm creation (monthly, region)	245.66	367.63	33.20	59.00	129.00	242.00	550.40	1.00	2256.00
COVID-19 deaths (per 100,000)	13.57	19.00	0.00	0.00	4.63	22.43	39.00	0.00	129.82
Population (thousands, region)	337.43	304.93	92.35	108.10	304.92	434.24	486.98	27.62	1202.73
Nightlight intensity (index)	18501.51	14543.96	0.00	0.00	20798.30	29768.59	35370.99	0.00	58158.60

Table A2: Within-Region Variation, Regression Window. Statistics are calculated on the March 2020 to November 2022 region-month regression sample. Each variable is first demeaned by region, so the table summarizes within-region month-to-month variation exploited by the fixed-effects models. Population is reported in thousands of persons.

Variable	Within-Region SD	Mean Abs. Deviation
SME openings (monthly, region)	182.58	87.86
SME closings (monthly, region)	13.06	8.11
SME net firm creation (monthly, region)	174.48	83.48
COVID-19 deaths (per 100,000)	18.66	14.33
Population (thousands, region)	3.98	2.64
Nightlight intensity (index)	12723.97	10145.24

Table A3: National Openings Rates by Firm Size and Pandemic Phase. National monthly openings rates per 1,000 lagged active firms, aggregated from the region-month panel. The 2019 baseline is the average over February–December 2019 because lagged active-stock denominators are required to construct rates. The first lockdown corresponds to March–May 2020.

Group	2019 baseline	First lockdown	2021 avg.	2022 avg.	Lockdown vs. 2019	2022 vs. 2019
All SMEs	14.11	6.12	11.74	16.21	-56.64%	14.91%
Small firms	14.19	6.15	11.80	16.32	-56.67%	15.00%
Medium firms	6.43	3.00	5.34	3.49	-53.32%	-45.70%

Table A4: Classification of Economic Activities into Sector Categories. This table presents the classification scheme used to aggregate economic activities into broader sector categories for the panel regression analysis. The classification is based on NACE Rev. 2 codes.

Sector Category	Subsector	Economic Activities (NACE Rev. 2 codes)
Primary	Agriculture	Horticulture and animal husbandry (01), Forestry (02), Fishing and Aquaculture (03)
	Mining	Coal and lignite mining (05), Oil and gas extraction (06), Mining of metal ores (07), Other mining (08), Mining support services (09)
Manufacturing	Food/Beverages	Food products (10), Drink production (11), Tobacco products (12)
	Textiles/Leather	Textiles (13), Clothing (14), Leather products (15)
	Industrial	Wood and cork (16), Paper (17), Printing (18), Petroleum products (19), Chemicals (20), Pharmaceuticals (21), Rubber and plastics (22), Non-metallic minerals (23), Basic metals (24), Metal products (25)
	Technology	Computers and electronics (26), Electrical equipment (27), Machinery (28), Motor vehicles (29), Other transport equipment (30)
	Other Manufacturing	Furniture (31), Other goods (32), Repair and installation (33)
Utilities	–	Electricity, gas, and air conditioning (35), Water collection and distribution (36), Sewage (37), Waste collection and disposal (38), Decontamination services (39)
Construction	–	Buildings (41), Civil engineering (42), Specialized construction (43)
Trade & Transport	Trade	Wholesale and retail of vehicles (45), Wholesale trade (46), Retail trade (47)
	Transport	Land transport (49), Water transport (50), Air transport (51), Warehousing (52), Postal services (53)
Services	Hospitality	Accommodation (55), Food and beverage service (56)
	Information/Media	Publishing (58), Film and television (59), Broadcasting (60), Telecommunications (61), Computer programming (62), Information services (63)
	Financial Professional	Financial services (64), Insurance (65), Financial support activities (66) Real estate (68), Legal and accounting (69), Management consulting (70), Engineering activities (71), Scientific research (72), Advertising and market research (73), Other professional activities (74), Veterinary activities (75)
	Business Support	Rental and leasing (77), Employment activities (78), Travel agencies (79), Security activities (80), Building services (81), Office administration (82)
Public & Social	–	Public administration (84), Education (85), Health activities (86), Residential care (87), Social services (88), Creative and entertainment (90), Libraries and museums (91), Gambling (92), Sports and recreation (93), Membership organizations (94), Repair of goods (95), Other personal services (96)
Other	–	Unclassified economic activities (99)

Table A5: Dynamic Rate Fixed-Effects Estimates without Sector Composition Controls

	(1)	(2)	(3)
	Openings rate	Net firm creation rate	Closings rate
Deaths (per 100k) $D_{r,t}$	-0.057 (0.052)	-0.052 (0.049)	-0.005 (0.004)
Deaths, lag 1 $D_{r,t-1}$	-0.013 (0.030)	-0.013 (0.030)	-0.000 (0.002)
Deaths, lag 2 $D_{r,t-2}$	-0.022 (0.016)	-0.019 (0.016)	-0.003 (0.002)
Deaths, lag 3 $D_{r,t-3}$	0.005 (0.036)	0.005 (0.036)	-0.000 (0.002)
Four-month cumulative effect	-0.087 (0.108) [$p=0.419$]	-0.079 (0.104) [$p=0.448$]	-0.008 (0.006) [$p=0.213$]
Wild-bootstrap p for $D_{r,t}$	[0.569]	[0.531]	[0.669]
Nightlight intensity	0.00005 (0.00004)	0.00006 (0.00004)	-0.00000 (0.00000)
Population	0.00026 (0.00020)	0.00026 (0.00019)	-0.00001 (0.00001)
Observations	363	363	363
Region FE (μ_r)	Yes	Yes	Yes
Month FE (μ_t)	Yes	Yes	Yes
SEs clustered by region	Yes	Yes	Yes

Notes: Fixed-effects regressions of SME opening, net-firm-creation, and closing rates, each measured per 1,000 lagged active SMEs, on the March 2020 to November 2022 sample. Region and month-by-date fixed effects are included. Robust standard errors clustered at the region level are in parentheses. Brackets report p -values for four-month cumulative-effect tests and wild-cluster bootstrap p -values for the contemporaneous mortality term. Deaths variables are measured per 100,000 population, and missing mortality months within the estimation window are coded as zero before constructing lags. This table omits sector-share controls and is reported as a simpler comparison to the main-text specification in Table 5.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Size-Split Net Firm Creation and Closings: Appendix Summary

Outcome	Size	$D_{r,t}$	Clustered p	Wild-bootstrap p	Four-month cumulative
Net firm creation rate	Small	-0.034**	0.031	0.147	-0.041 [$p=0.288$]
Net firm creation rate	Medium	-0.195**	0.011	0.029	0.220 [$p=0.303$]
Closings rate	Small	-0.003	0.278	0.567	-0.006 [$p=0.134$]
Closings rate	Medium	n.a.	n.a.	n.a.	n.a.

Notes: This table summarizes sector-controlled size-split rate regressions for net firm creation and closings over the March 2020 to November 2022 region-month sample. All specifications include region and month-by-date fixed effects, nightlight intensity, population, four mortality lags, and the outcome-specific regional sector-composition controls used in the main text. Wild-bootstrap p -values refer to the contemporaneous mortality coefficient. Medium-firm closings are not estimable because the underlying medium-closing series is zero throughout the estimation window.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Placebo Leads in the Small-Firm Entry Specification

	Estimate	Clustered p	Wild-bootstrap p
Deaths, lead 1 $D_{r,t+1}$	-0.049	0.212	0.414
Deaths, lead 2 $D_{r,t+2}$	0.013	0.434	0.423
Deaths, lead 3 $D_{r,t+3}$	-0.017	0.584	0.601
Joint test of leads	$[p=0.157]$		

Notes: The specification matches the main small-firm openings-rate regression in Table 7 but adds future mortality leads. The sample is March 2020 to November 2022. The joint test reports the p -value from a Wald test that the three lead coefficients are jointly zero.

Table A8: Shifted-Mortality Pre-COVID Placebo for Small-Firm Entry

	Estimate	p -value
Shifted mortality term	0.003	0.930
Wild-bootstrap p	$[0.931]$	
Four-month cumulative placebo effect	2.013	0.211
Observations	99	

Notes: This placebo re-estimates the main small-firm openings-rate specification on the pre-COVID window from March to November 2019. The later mortality path is shifted backward by 12 months and assigned to this earlier period as a fake treatment. The absence of a meaningful placebo association is reassuring for the interpretation of the main small-firm entry result.

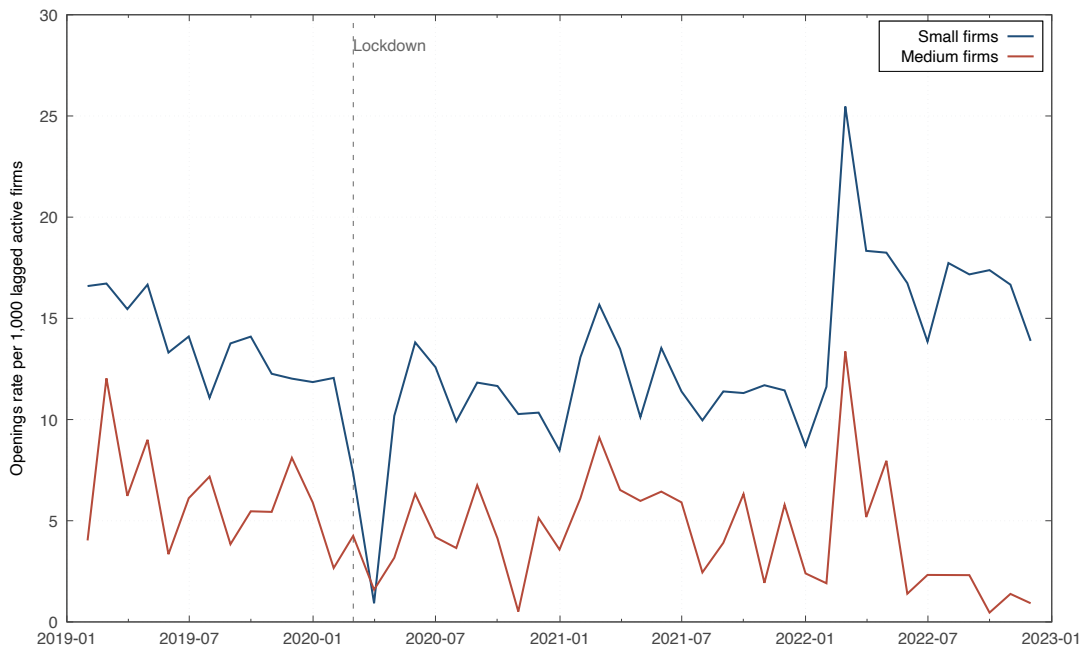
Figure A1: Monthly Openings Rates by Firm Size, 2019–2022. National monthly openings rates per 1,000 lagged active firms for small and medium enterprises. The vertical dashed line marks March 2020, the onset of the first lockdown period used in the paper’s descriptive comparisons.

Figure A2: Small-Firm Openings by Sector, 2019–2022. National monthly totals of small-firm openings in the four broad sectors most relevant to the paper’s heterogeneity discussion. The vertical dashed line marks March 2020.

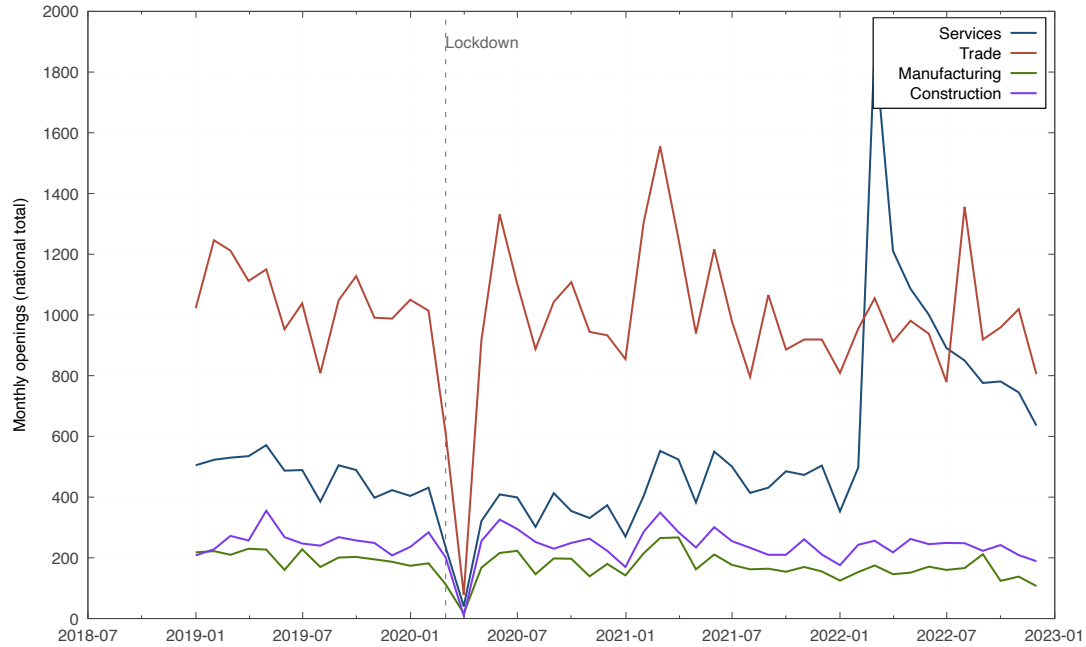
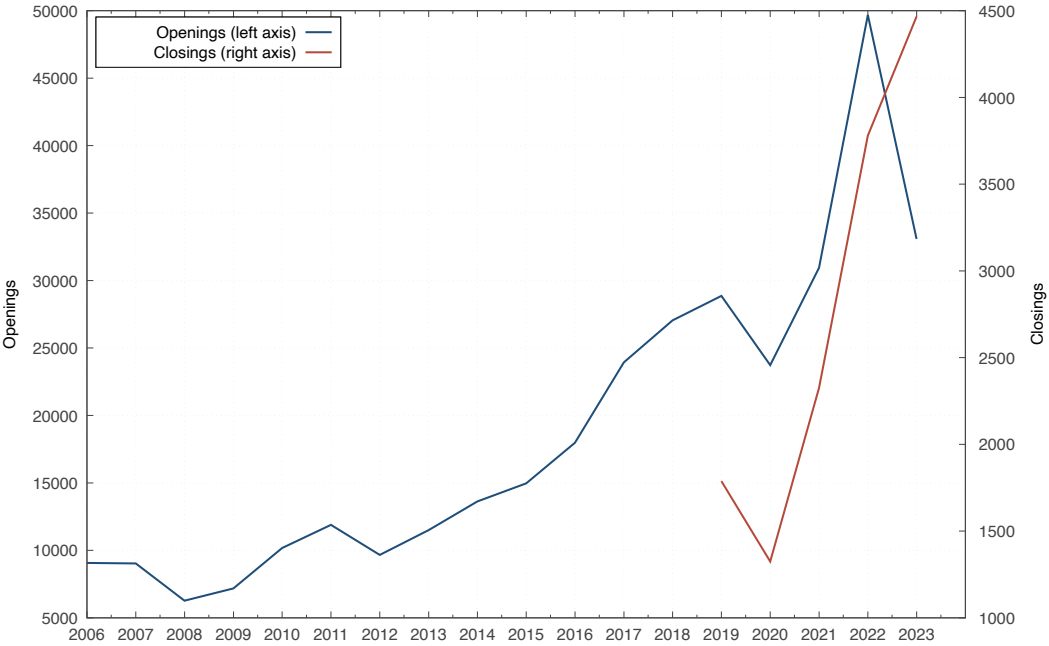


Figure A3: Monthly Trends in Company Openings and Closings, 2019–2022. Monthly counts of company openings and closings recorded from 2019 to 2022. Openings are read on the left axis and closings on the right axis. Data sourced from the Business Register maintained by the National Statistics Office of Georgia. The figure highlights the impact of economic changes and policies over this period.



Figure A4: Annual Trends in Company Openings (2006–2023) and Closings (2019–2023) in Georgia. The figure illustrates the evolution of company dynamics in Georgia, showing the annual number of new registrations (2006–2023) and company closures (2019–2023). Openings are read on the left axis and closings on the right axis. Data are drawn from the Business Register maintained by the National Statistics Office of Georgia. Information on closings is available from 2019 onward, as it covers active companies registered in 2018 or earlier that subsequently ceased operations.



Appendix B Proofs

Proof of Proposition 1

At the period-1 cutoff,

$$W_1(z_E^1) = F. \quad (29)$$

Consider a period-0 entrant with productivity z_E^1 . Its entry value is

$$\begin{aligned} J_0(z_E^1) &= \pi_0^*(z_E^1) + \beta F - F \\ &= \Omega A_0^\sigma (z_E^1)^\sigma - c - (1 - \beta)F. \end{aligned} \quad (30)$$

Using (19),

$$J_0(z_E^1) > 0 \iff A_0^\sigma > (1 - \beta)A_1^\sigma + \beta. \quad (31)$$

Condition (E) therefore implies that a firm indifferent about entering in period 1 strictly prefers entering in period 0. Since entry values are increasing in z , $z_E^0 < z_E^1$. The pool of potential entrants is the same in both periods, so the higher cutoff implies fewer entrants in period 1. Finally, optimal employment in (14) is increasing in z , which establishes the size result. $\square$

Proof of Proposition 2

Define

$$H_0 = \frac{(1 - \gamma)A_0^{\frac{1}{1-\gamma}} - \alpha}{1 - \alpha - \gamma}, \quad D_1 = (1 - \beta)A_1^\sigma + \beta.$$

The period-1 operating cutoff is

$$z_X^1 = \left(\frac{c}{\Omega D_1} \right)^{1/\sigma}.$$

For firms with positive period-1 continuation value, period-0 value can be written as

$$(1 - \beta)W_0^I(z) = [(1 - \beta)H_0 + \beta D_1] \Omega z^\sigma - c.$$

Let $Q_0 = (1 - \beta)H_0 + \beta D_1$. In the nondegenerate case $Q_0 > 0$, the period-0 operating cutoff is

$$z_X^0 = \left(\frac{c}{\Omega Q_0} \right)^{1/\sigma}.$$

Condition (S), $H_0 < D_1$, implies $Q_0 < D_1$ and therefore $z_X^0 > z_X^1$. Moreover, if $z < z_X^1$, then

$$D_1 \Omega z^\sigma - c < 0.$$

Condition (S) consequently implies

$$H_0\Omega z^\sigma - c < D_1\Omega z^\sigma - c < 0.$$

Hence, no firm below z_X^1 operates in period 0 with the intention of exiting before period 1. Every incumbent operating in period 0 therefore has

$$z \geq z_X^0 > z_X^1$$

and continues in period 1. $\square$